Project–High Level Design On

Marketing Content Tool

Course Name: Agentic AI

Institution Name: Medicaps University–Datagami Skill Based Course

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Academic Year: 2026

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1. Introduction

This project is based on the concept of Agentic AI, which refers to intelligent systems that can perform tasks independently, make decisions, and achieve specific goals. In today's digital era, businesses rely heavily on marketing content to attract and engage customers, but creating and managing such content requires time and effort. This project aims to develop an Agentic AI-based marketing content tool that can automatically generate and optimize content. By using multiple AI agents, the system can handle tasks like content creation, improvement, and analysis, making the overall process more efficient and effective.

1.1 Scope of the Document

This document provides an overview of the proposed system, including its purpose, features, and overall design. It explains how the Agentic AI-based marketing tool works and highlights its main components. The document focuses on the high-level design and does not include detailed technical implementation.

1.2 Intended Audience

- Students (for academic understanding and project reference)
- Project evaluators and faculty members
- Developers who want to understand the system design
- Beginners interested in Agentic AI concepts
- Anyone looking to learn about AI-based marketing tools

2. System Overview

The proposed system is an Agentic AI-based marketing content tool designed to automate and improve the process of creating marketing content. The system uses multiple intelligent agents that work together to perform different tasks such as content generation, content optimization, audience targeting, and performance analysis.

The user provides input such as marketing goals, target audience, or type of content required. Based on this input, the system processes the request through an agent orchestrator, which assigns tasks to different AI agents. Each agent performs its specific function and collaborates with other agents to generate high-quality output.

The system also integrates with advanced language models to produce relevant and engaging content. The final output can include social media posts, blog content, or advertisement copies. Overall, the system aims to reduce manual effort, save time, and provide efficient and effective marketing solutions.

2.1 System Design

The system design is based on an Agentic AI architecture where different components work together to perform tasks efficiently. It includes the user interface, backend system, AI agents, and data storage.

The user provides input through the interface, which is processed by the backend using an API gateway and agent orchestrator. The orchestrator manages the workflow and assigns tasks to different AI agents.

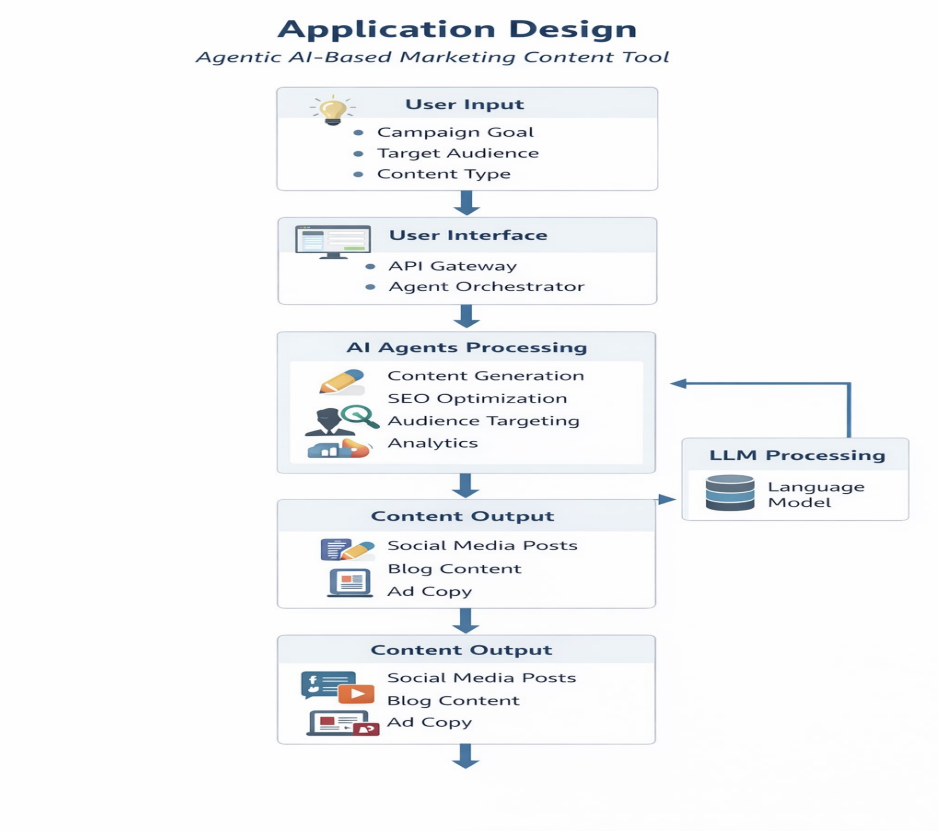
These agents handle tasks like content generation, optimization, and analysis using a language model. All components work together to generate and deliver effective marketing content in an efficient manner.

2.2 Application Design

The application is designed using an Agentic AI approach where multiple intelligent agents work together to perform different tasks. The system follows a simple modular structure with components like user interface, backend system, and AI agents.

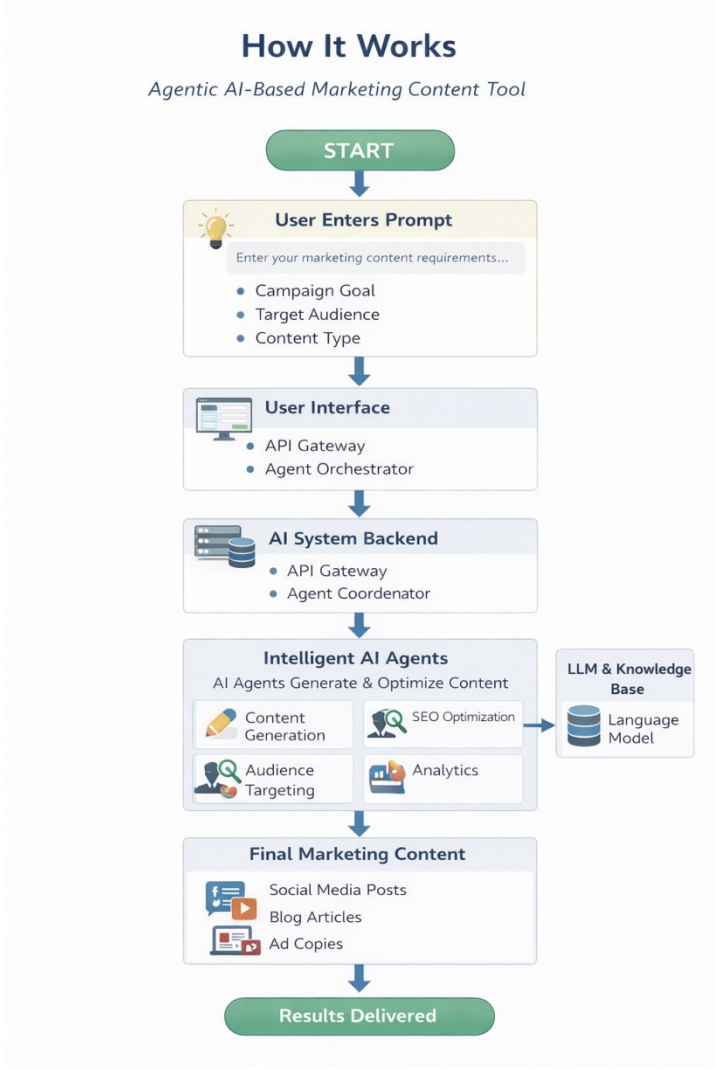
The user provides input such as campaign goals and target audience, which is processed by the backend through an API gateway and agent orchestrator. The orchestrator assigns tasks to different agents like content generation, SEO optimization, and analytics.

These agents use a language model to generate and improve content. Finally, the system produces output in the form of social media posts, blog content, and ad copies, making the process faster and more efficient.



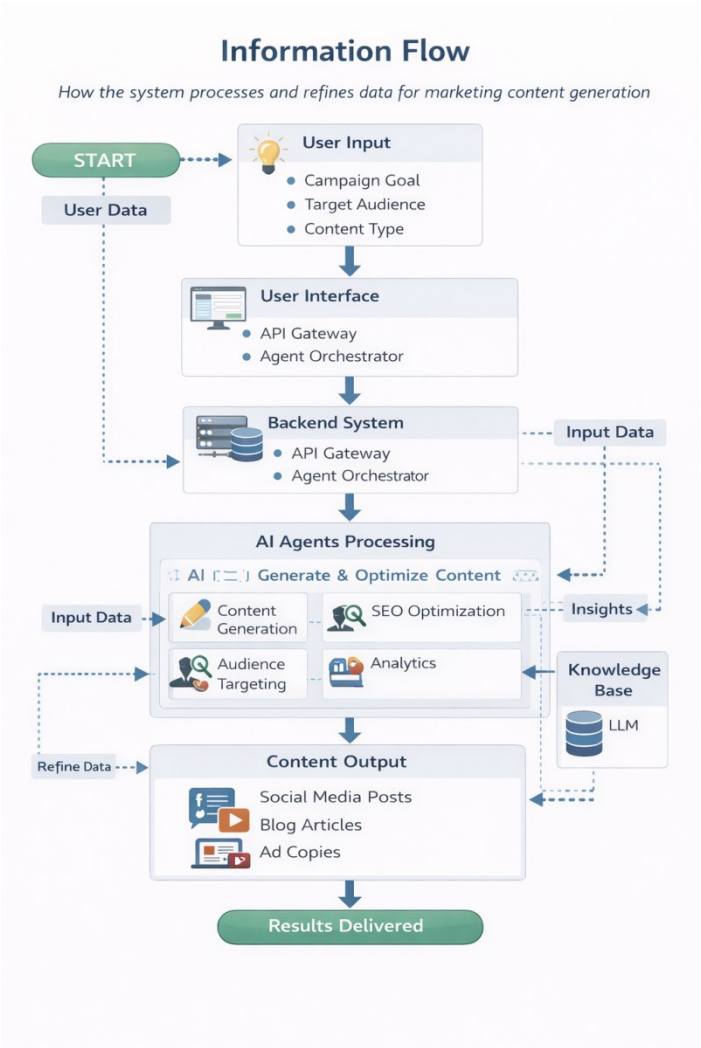
2.3 Process Flow

The process starts when the user enters a prompt by providing details like campaign goal, target audience, and content type. This input is processed by the system through the backend, where the agent orchestrator assigns tasks to different AI agents. These agents generate, optimize, and refine the content using a language model. Finally, the system produces the required marketing content such as social media posts, blogs, or ad copies and delivers it to the user.



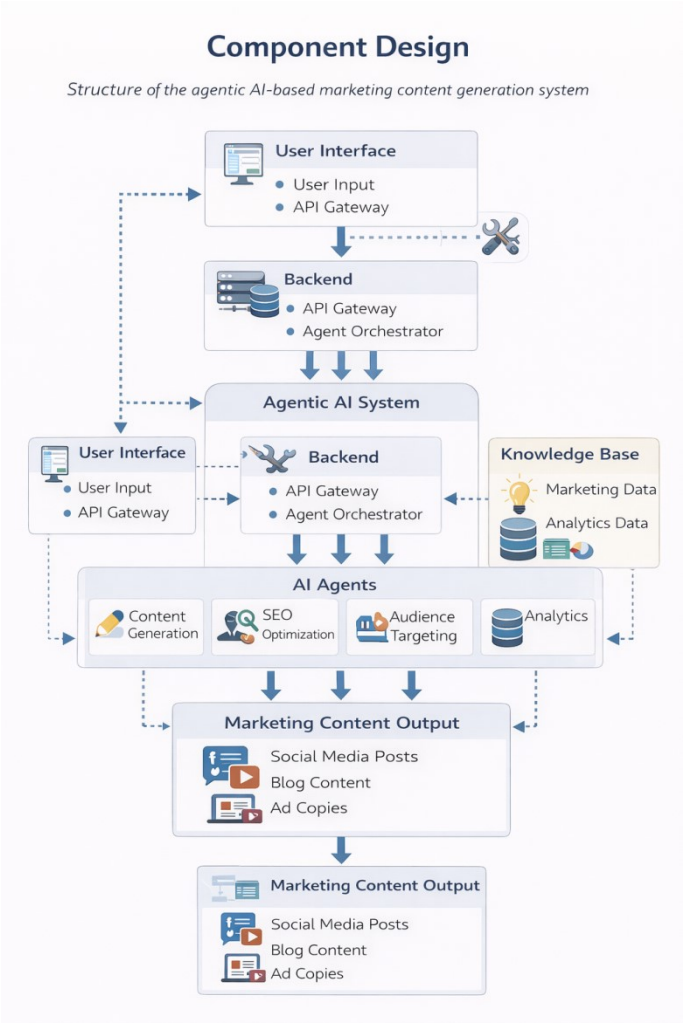
2.4 Information Flow

The information flow in the system begins with the user input, which includes details like campaign goals, target audience, and content type. This data is passed through the user interface to the backend system, where it is processed by the API gateway and agent orchestrator. The information is then distributed to different AI agents, which generate, optimize, and analyze the content using a language model and knowledge base. The processed data flows back as refined content, which is finally delivered to the user in the form of marketing outputs such as social media posts, blogs, and ad copies.



2.5 Component Design

The component design of the system includes major parts such as the user interface, backend system, AI agents, and knowledge base. The user interface collects input from the user and sends it to the backend through the API gateway. The backend, managed by the agent orchestrator, controls the overall workflow and task distribution.The AI agents are responsible for performing specific tasks like content generation, SEO optimization, audience targeting, and analytics. These agents use the knowledge base and language model to generate and refine content. Finally, all components work together to produce and deliver the final marketing content efficiently.



2.6 Key Design Considerations

- **Scalability:**
  The system should be able to handle an increasing number of users and requests without affecting performance. As more users use the platform, the architecture should support easy expansion.
- **Performance:**
  Fast response time is important so that users can quickly get generated content. The system should process inputs efficiently and minimize delays.
- **Modularity:**
  The application is divided into different components such as user interface, backend, and AI agents. This makes the system easier to maintain, update, and improve without affecting the entire system.
- **Accuracy and Quality:**
  The generated marketing content should be relevant, meaningful, and aligned with the user's requirements. The system should ensure high-quality outputs.
- **Security:**
  User data, inputs, and generated content should be protected. Proper authentication and data handling methods should be used to avoid misuse.
- **Flexibility:**
  The system should support different types of content like blogs, social media posts, and advertisements. It should also adapt to different marketing needs.
- **Reliability:**
  The system should work consistently without errors or crashes. It should provide stable performance even under heavy usage.

2.7 API Catalogue

The API catalogue defines the different APIs used in the system to enable communication between various components and external services. Each API performs a specific function to ensure smooth processing and efficient content generation.

1. Content Generation API

This API is responsible for generating marketing content based on user input. It interacts with the language model to create relevant and high-quality content.

- **Request Parameters:** campaignGoal, targetAudience, contentType, keywords (optional)
- **Response:** generatedContent, contentType, status (success/failure)

2. Regenerate Content API
This API allows users to modify or improve already generated content based on new instructions or preferences. It helps in refining the output as per user needs.

- **Request Parameters:** existingContent, modificationInstructions, tone/style (optional)
- **Response:** regeneratedContent, changesApplied, status

3. User Input Validation API
This API ensures that the input provided by the user is correct and complete before processing. It helps in avoiding errors and improving system reliability.

- **Request Parameters:** userInputData (campaignGoal, targetAudience, contentType)
- **Response:** validationStatus (valid/invalid), errorMessage (if any)

4. External API (Integration)
This API is used to connect the system with external platforms such as social media or analytics tools. It enables content publishing and performance tracking.

- **Key Parameters:** platformName, apiKey/authenticationToken, contentData
- **Response:** publishingStatus, platformResponse, errorDetails (if any)

3. Data Design

The data design of the proposed system focuses on how data is structured, stored, accessed, and managed throughout the lifecycle of the application. Since the system deals with multiple types of data such as user inputs, generated marketing content, and analytical results, it is important to maintain a well-organized and efficient data structure. A proper data design ensures smooth data flow between components, faster processing, and improved system performance. It also helps in maintaining data consistency, security, and scalability as the number of users and data increases over time.

3.1 Data Model

The data model defines the logical structure of data used in the system. It represents how different data entities are organized and how they are related to each other. The main entities in this system include:

- **User Entity:**
  Stores user-related information such as user ID, name, login credentials, and preferences.
- **Input Entity:**
  Contains details provided by the user such as campaign goals, target audience, content type, and keywords.
- **Content Entity:**
  Stores the generated and regenerated marketing content including blogs, social media posts, and advertisement copies.
- **Analytics Entity:**
  Maintains data related to performance analysis such as engagement metrics, predicted reach, and effectiveness of the content.

- **Relationships:**
  A user provides multiple inputs → each input generates content → each content can have associated analytics.

This structured model ensures that all data is interconnected and can be easily managed and retrieved.

3.2 Data Access Mechanism

The data access mechanism describes how data is stored, retrieved, updated, and managed within the system. The backend system interacts with the database through APIs to ensure smooth and secure data handling.

- The system uses databases such as SQL (structured) or NoSQL (flexible) depending on data requirements.
- APIs are used to perform operations like create, read, update, and delete (CRUD).
- Efficient querying methods are implemented to ensure fast data retrieval.
- Authentication and authorization techniques are used to restrict unauthorized access.
- Data caching may be used to improve performance and reduce response time.

This mechanism ensures that data operations are performed efficiently and securely without affecting system performance.

3.3 Data Retention Policies

Data retention policies define how long data is stored in the system and how it is managed over time. These policies are important for maintaining system efficiency, reducing storage load, and ensuring data privacy.

- User data is stored only for the duration required for system functionality.
- Temporary or unused data is periodically deleted to free up storage.
- Important data such as generated content and analytics may be stored for future reference.
- Regular backups are maintained to prevent data loss.
- Sensitive information is encrypted and handled securely.

These policies help in balancing data availability with storage efficiency and security requirements.

3.4 Data Migration

Data migration refers to the process of transferring data from one system, database, or format to another. This is important during system upgrades, scaling, or when changing technologies.

- Data is migrated in a structured manner to avoid loss or corruption.
- Backup of existing data is taken before starting the migration process.
- Data validation is performed after migration to ensure accuracy and completeness.
- Automated tools or scripts may be used to simplify the migration process.
- Migration is planned carefully to minimize system downtime.

Proper data migration ensures that system updates or changes can be implemented smoothly without affecting existing data or system functionality.

4. Interfaces

The interface design of the system defines how different components interact with each other and how users communicate with the application. Proper interface design ensures smooth data flow, better usability, and efficient system performance.

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1. User Interface

The user interface (UI) is the front-end part of the system through which users interact with the application. It is designed to be simple, user-friendly, and easy to navigate.

- Allows users to enter inputs such as campaign goals, target audience, and content type
- Displays generated content in a clear and readable format
- Provides options to regenerate or modify content
- Includes basic navigation and dashboard features
- Designed to be responsive and easy to use

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2. API Interface

The API interface enables communication between the frontend and backend systems. It ensures that user requests are properly sent and processed.

- Handles sending and receiving data between components
- Supports different APIs like content generation and validation
- Uses standard protocols for communication (e.g., HTTP/REST)
- Ensures secure and reliable data transfer

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3. Backend AI Server Interface

This interface connects the backend system with the AI models and agents. It is responsible for processing requests and generating intelligent outputs.

- Communicates with AI agents and language models
- Processes input data and generates content
- Manages task execution through agent orchestrator
- Ensures efficient handling of AI operations

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4. Backend Database Interface

The backend database interface manages the interaction between the backend system and the database.

- Stores user data, inputs, and generated content
- Retrieves data when required for processing
- Supports CRUD operations (create, read, update, delete)
- Ensures secure and efficient data handling

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5. External Interfaces

External interfaces allow the system to connect with third-party platforms and services.

- Integration with social media platforms for content publishing
- Connection with analytics tools for performance tracking
- Uses external APIs for extended functionality
- Ensures smooth data exchange with outside systems

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6. Interface Design Considerations

The design of interfaces is important to ensure usability and system efficiency.

- **Usability:** Interfaces should be simple and easy to understand
- **Consistency:** Maintain uniform design across all interfaces
- **Security:** Protect data during communication
- **Performance:** Ensure fast and smooth data transfer
- **Scalability:** Support future expansion and integration
- **Reliability:** Interfaces should work without errors or failures

Overall, well-designed interfaces help in improving user experience and system performance.

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5. State and Session Management

State and session management play a crucial role in maintaining user interactions and ensuring smooth functioning of the system. Since the proposed application is based on an Agentic AI architecture and follows a web-based model, it is designed to be mostly stateless while still maintaining necessary session information for users.

1. Stateless Nature of the System

The system follows a stateless architecture at the API level, meaning each request from the user contains all the required information and does not depend on previous requests. This improves scalability and simplifies backend processing. However, certain user-specific data is maintained separately using session management techniques.

2. Session Management

Session management is used to track user interactions across multiple requests. When a user starts using the system, a unique session ID is generated and associated with that user.

- Session ID is used to identify and track user activity
- Sessions help in storing temporary data like recent inputs or generated content

- Session timeout is implemented to automatically end inactive sessions
- Sessions can be managed using tokens (e.g., JWT) or server-side storage

3. State Management

State management refers to maintaining the current status of user interactions within the system.

- Tracks user inputs, generated content, and progress
- Helps in resuming tasks like content regeneration
- Maintains consistency across multiple steps of processing
- Can be handled using frontend state (UI) and backend storage

4. Handling User Sessions

Efficient handling of user sessions ensures smooth user experience.

- Each user session is uniquely identified
- Session data is stored securely
- Multiple sessions can be handled simultaneously
- Session data is cleared after logout or timeout

5. Scalability Considerations

- Stateless APIs allow easy scaling of backend servers
- Load balancing can be used to distribute user requests
- Session storage can be centralized (e.g., Redis) for large-scale systems
- System should support multiple concurrent users efficiently

6. Security Considerations

- Secure session IDs and tokens to prevent misuse
- Use encryption for sensitive session data
- Implement authentication and authorization mechanisms
- Protect against session hijacking and unauthorized access
- Use HTTPS for secure communication

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6. Caching

Caching is used to improve system performance by storing frequently accessed data temporarily so that it can be retrieved quickly without repeated processing. In an Agentic AI system, caching plays an important role in reducing response time and optimizing resource usage.

1. Purpose of Caching

- Reduce repeated API calls to AI models
- Improve response time for users
- Minimize system load and cost

2. Types of Data Cached

- Frequently generated content
- User session data (temporary)
- API responses and results
- Static data like templates or configurations

3. Caching Mechanism

- In-memory caching systems such as Redis or Memcached can be used
- Data is stored with a unique key for quick retrieval
- Time-to-Live (TTL) is set to automatically expire outdated data

4. Benefits of Caching

- Faster response time
- Reduced latency in content generation
- Improved overall system performance
- Better user experience

5. Cache Invalidation

- Old or outdated data is removed after a fixed time
- Cache is refreshed when new data is generated
- Ensures that users always receive updated information

6. Scalability and Performance

- Caching reduces load on backend and AI models
- Supports handling of large number of requests
- Improves system efficiency during peak usage

7. Security Considerations

- Avoid storing sensitive data in cache
- Use secure access control for cached data
- Ensure proper data expiration to prevent misuse

Overall, caching enhances the efficiency and speed of the system while maintaining optimal resource usage.

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7. Non-Functional Requirements

Non-functional requirements define the quality attributes of the system such as security, performance, reliability, and usability. These requirements ensure that the system not only works correctly but also performs efficiently and securely under different conditions.

7.1 Security Aspects

Security is an important aspect of the system as it handles user data and generated content. Proper measures must be taken to protect the system from unauthorized access and data breaches.

- **Authentication and Authorization:**
  Only authorized users should be able to access the system. Login mechanisms and role-based access control can be used.
- **Data Encryption:**
  Sensitive data such as user inputs and session information should be encrypted during storage and transmission.

- **Secure APIs:**
  All APIs should be protected using secure protocols (HTTPS) and authentication tokens.
- **Protection Against Attacks:**
  The system should be protected from common threats like SQL injection, cross-site scripting (XSS), and session hijacking.
- **Data Privacy:**
  User data should be handled carefully and not shared without permission.
- **Session Security:**
  Secure session handling should be implemented to prevent unauthorized access.

7.2 Performance Aspects

Performance defines how efficiently the system responds to user requests and handles workloads.

- **Response Time:**
  The system should generate content quickly with minimal delay.
- **Scalability:**
  The system should handle increasing number of users and requests without performance degradation.
- **Throughput:**
  The system should be able to process multiple requests simultaneously.
- **Resource Optimization:**
  Efficient use of CPU, memory, and network resources should be ensured.
- **Load Handling:**
  The system should perform well under high traffic conditions.
- **Caching:**
  Use of caching techniques to reduce repeated processing and improve speed.

Overall, these non-functional requirements ensure that the system is secure, reliable, and performs efficiently in real-world conditions.

8. References

The following references were used for understanding the concepts of Agentic AI, Artificial Intelligence in marketing, and system design:

1. OpenAI – *Practices for Governing Agentic AI Systems*
   OpenAI Agentic AI Paper

2. ScienceDirect – *Artificial Intelligence (AI) Applications for Marketing*
   AI in Marketing Research Paper

3. Springer – *Agentic AI: Architectures, Applications, and Future Directions*
   Agentic AI Survey Paper

4. MDPI – *Agentic AI and Large Language Models Research Study*
   Agentic AI Research Landscape

5. Springer – *AI in Marketing Framework Study*
   AI Marketing Framework Paper

6. Agentic AI Documentation – *Agentic AI Knowledge Base*
   Agentic AI Documentation

These references helped in understanding the design, working, and application of Agentic AI in marketing content generation systems.